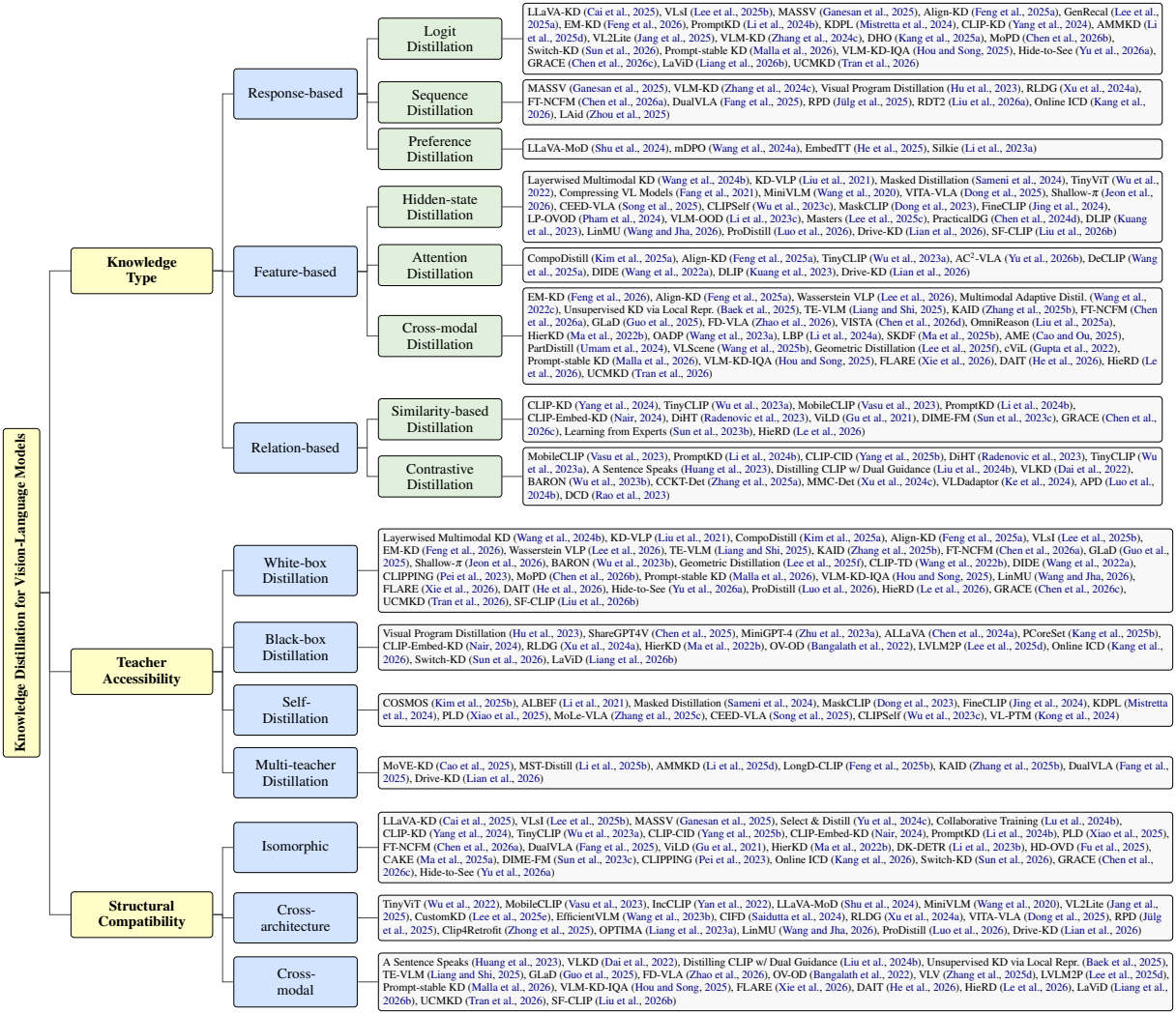

Knowledge Distillation for Vision-Language Models
Knowledge Type
Response-based
Logit Distillation
LLaVA-KD (Cai et al., 2025), VLsI (Lee et al., 2025b), MASSV (Ganesan et al., 2025), Align-KD (Feng et al., 2025a), GenRecal (Lee et al., 2025a), EM-KD (Feng et al., 2026), PromptKD (Li et al., 2024b), KDPL (Mistretta et al., 2024), CLIP-KD (Yang et al., 2024), AMMKD (Li et al., 2025d), VL2Lite (Jang et al., 2025), VLM-KD (Zhang et al., 2024c), DHO (Kang et al., 2025a), MoPD (Chen et al., 2026b), Switch-KD (Sun et al., 2026), Prompt-stable KD (Malla et al., 2026), VLM-KD-IQA (Hou and Song, 2025), Hide-to-See (Yu et al., 2026a), GRACE (Chen et al., 2026c), LaViD (Liang et al., 2026b), UCMKD (Tran et al., 2026)
Sequence Distillation
MASSV (Ganesan et al., 2025), VLM-KD (Zhang et al., 2024c), Visual Program Distillation (Hu et al., 2023), RLDG (Xu et al., 2024a), FT-NCFM (Chen et al., 2026a), DualVLA (Fang et al., 2025), RPD (Jülg et al., 2025), RDT2 (Liu et al., 2026a), Online ICD (Kang et al., 2026), LAid (Zhou et al., 2025)
Preference Distillation
LLaVA-MoD (Shu et al., 2024), mDPO (Wang et al., 2024a), EmbedTT (He et al., 2025), Silkie (Li et al., 2023a)
Feature-based
Hidden-state Distillation
Layerwised Multimodal KD (Wang et al., 2024b), KD-VLP (Liu et al., 2021), Masked Distillation (Sameni et al., 2024), TinyViT (Wu et al., 2022), Compressing VL Models (Fang et al., 2021), MiniVLM (Wang et al., 2020), VITA-VLA (Dong et al., 2025), Shallow-π (Jeon et al., 2026), CEED-VLA (Song et al., 2025), CLIPSelf (Wu et al., 2023c), MaskCLIP (Dong et al., 2023), FineCLIP (Jing et al., 2024), LP-OVOD (Pham et al., 2024), VLM-OOD (Li et al., 2023c), Masters (Lee et al., 2025c), PracticalDG (Chen et al., 2024d), DLIP (Kuang et al., 2023), LinMU (Wang and Jha, 2026), ProDistill (Luo et al., 2026), Drive-KD (Lian et al., 2026), SF-CLIP (Liu et al., 2026b)
Attention Distillation
CompoDistill (Kim et al., 2025a), Align-KD (Feng et al., 2025a), TinyCLIP (Wu et al., 2023a), AC^2-VLA (Yu et al., 2026b), DeCLIP (Wang et al., 2025a), DIDE (Wang et al., 2022a), DLIP (Kuang et al., 2023), Drive-KD (Lian et al., 2026)
Cross-modal Distillation
EM-KD (Feng et al., 2026), Align-KD (Feng et al., 2025a), Wasserstein VLP (Lee et al., 2026), Multimodal Adaptive Distil. (Wang et al., 2022c), Unsupervised KD via Local Repr. (Baek et al., 2025), TE-VLM (Liang and Shi, 2025), KAID (Zhang et al., 2025b), FT-NCFM (Chen et al., 2026a), GLaD (Guo et al., 2025), FD-VLA (Zhao et al., 2026), VISTA (Chen et al., 2026d), OmniReason (Liu et al., 2025a), HierKD (Ma et al., 2022b), OADP (Wang et al., 2023a), LBP (Li et al., 2024a), SKDF (Ma et al., 2025b), AME (Cao and Ou, 2025), PartDistill (Umam et al., 2024), VLScene (Wang et al., 2025b), Geometric Distillation (Lee et al., 2025f), cViL (Gupta et al., 2022), Prompt-stable KD (Malla et al., 2026), VLM-KD-IQA (Hou and Song, 2025), FLARE (Xie et al., 2026), DAIT (He et al., 2026), HieRD (Le et al., 2026), UCMKD (Tran et al., 2026)
Relation-based
Similarity-based Distillation
CLIP-KD (Yang et al., 2024), TinyCLIP (Wu et al., 2023a), MobileCLIP (Vasu et al., 2023), PromptKD (Li et al., 2024b), CLIP-Embed-KD (Nair, 2024), DiHT (Radenovic et al., 2023), ViLD (Gu et al., 2021), DIME-FM (Sun et al., 2023c), GRACE (Chen et al., 2026c), Learning from Experts (Sun et al., 2023b), HieRD (Le et al., 2026)
Contrastive Distillation
MobileCLIP (Vasu et al., 2023), PromptKD (Li et al., 2024b), CLIP-CID (Yang et al., 2025b), DiHT (Radenovic et al., 2023), TinyCLIP (Wu et al., 2023a), A Sentence Speaks (Huang et al., 2023), Distilling CLIP w/ Dual Guidance (Liu et al., 2024b), VLKD (Dai et al., 2022), BARON (Wu et al., 2023b), CCKT-Det (Zhang et al., 2025a), MMC-Det (Xu et al., 2024c), VLDadaptor (Ke et al., 2024), APD (Luo et al., 2024b), DCD (Rao et al., 2023)
Teacher Accessibility
White-box Distillation
Layerwised Multimodal KD (Wang et al., 2024b), KD-VLP (Liu et al., 2021), CompoDistill (Kim et al., 2025a), Align-KD (Feng et al., 2025a), VLsI (Lee et al., 2025b), EM-KD (Feng et al., 2026), Wasserstein VLP (Lee et al., 2026), TE-VLM (Liang and Shi, 2025), KAID (Zhang et al., 2025b), FT-NCFM (Chen et al., 2026a), GLaD (Guo et al., 2025), Shallow-π (Jeon et al., 2026), BARON (Wu et al., 2023b), Geometric Distillation (Lee et al., 2025f), CLIP-TD (Wang et al., 2022b), DIDE (Wang et al., 2022a), CLIPPING (Pei et al., 2023), MoPD (Chen et al., 2026b), Prompt-stable KD (Malla et al., 2026), VLM-KD-IQA (Hou and Song, 2025), LinMU (Wang and Jha, 2026), FLARE (Xie et al., 2026), DAIT (He et al., 2026), Hide-to-See (Yu et al., 2026a), ProDistill (Luo et al., 2026), HieRD (Le et al., 2026), GRACE (Chen et al., 2026c), UCMKD (Tran et al., 2026), SF-CLIP (Liu et al., 2026b)
Black-box Distillation
Visual Program Distillation (Hu et al., 2023), ShareGPT4V (Chen et al., 2025), MiniGPT-4 (Zhu et al., 2023a), ALLaVA (Chen et al., 2024a), PCoreSet (Kang et al., 2025b), CLIP-Embed-KD (Nair, 2024), RLDG (Xu et al., 2024a), HierKD (Ma et al., 2022b), OV-OD (Bangalath et al., 2022), LVLM2P (Lee et al., 2025d), Online ICD (Kang et al., 2026), Switch-KD (Sun et al., 2026), LaViD (Liang et al., 2026b)
Self-Distillation
COSMOS (Kim et al., 2025b), ALBEF (Li et al., 2021), Masked Distillation (Sameni et al., 2024), MaskCLIP (Dong et al., 2023), FineCLIP (Jing et al., 2024), KDPL (Mistretta et al., 2024), PLD (Xiao et al., 2025), MoLe-VLA (Zhang et al., 2025c), CEED-VLA (Song et al., 2025), CLIPSelf (Wu et al., 2023c), VL-PTM (Kong et al., 2024)
Multi-teacher Distillation
MoVE-KD (Cao et al., 2025), MST-Distill (Li et al., 2025b), AMMKD (Li et al., 2025d), LongD-CLIP (Feng et al., 2025b), KAID (Zhang et al., 2025b), DualVLA (Fang et al., 2025), Drive-KD (Lian et al., 2026)
Structural Compatibility
Isomorphic
LLaVA-KD (Cai et al., 2025), VLsI (Lee et al., 2025b), MASSV (Ganesan et al., 2025), Select & Distill (Yu et al., 2024c), Collaborative Training (Lu et al., 2024b), CLIP-KD (Yang et al., 2024), TinyCLIP (Wu et al., 2023a), CLIP-CID (Yang et al., 2025b), CLIP-Embed-KD (Nair, 2024), PromptKD (Li et al., 2024b), PLD (Xiao et al., 2025), FT-NCFM (Chen et al., 2026a), DualVLA (Fang et al., 2025), ViLD (Gu et al., 2021), HierKD (Ma et al., 2022b), DK-DETR (Li et al., 2023b), HD-OVD (Fu et al., 2025), CAKE (Ma et al., 2025a), DIME-FM (Sun et al., 2023c), CLIPPING (Pei et al., 2023), Online ICD (Kang et al., 2026), Switch-KD (Sun et al., 2026), GRACE (Chen et al., 2026c), Hide-to-See (Yu et al., 2026a)
Cross-architecture
TinyViT (Wu et al., 2022), MobileCLIP (Vasu et al., 2023), IncCLIP (Yan et al., 2022), LLaVA-MoD (Shu et al., 2024), MiniVLM (Wang et al., 2020), VL2Lite (Jang et al., 2025), CustomKD (Lee et al., 2025e), EfficientVLM (Wang et al., 2023b), CIFD (Saidutta et al., 2024), RLDG (Xu et al., 2024a), VITA-VLA (Dong et al., 2025), RPD (Jülg et al., 2025), Clip4Retrofit (Zhong et al., 2025), OPTIMA (Liang et al., 2023a), LinMU (Wang and Jha, 2026), ProDistill (Luo et al., 2026), Drive-KD (Lian et al., 2026)
Cross-modal
A Sentence Speaks (Huang et al., 2023), VLKD (Dai et al., 2022), Distilling CLIP w/ Dual Guidance (Liu et al., 2024b), Unsupervised KD via Local Repr. (Baek et al., 2025), TE-VLM (Liang and Shi, 2025), GLaD (Guo et al., 2025), FD-VLA (Zhao et al., 2026), OV-OD (Bangalath et al., 2022), VLV (Zhang et al., 2025d), LVLM2P (Lee et al., 2025d), Prompt-stable KD (Malla et al., 2026), VLM-KD-IQA (Hou and Song, 2025), FLARE (Xie et al., 2026), DAIT (He et al., 2026), HieRD (Le et al., 2026), LaViD (Liang et al., 2026b), UCMKD (Tran et al., 2026), SF-CLIP (Liu et al., 2026b)